

Statistics I

CSIT — 2nd Semester — 2080 — Answer Sheet
Subject Code: STA164

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Define statistics and discuss its importance in the field of computational sciences. The following are the numbers of minutes that a person had to wait for the bus to work on 20 working days: 15, 10, 2, 17, 5, 8, 3, 10, 2, 9, 8, 9, 13, 1, 10, 12, 5, 10, 8, 4. Compute mean, median, mode, standard deviation, variance and coefficient of variation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Define normal distribution. What are the main characteristics of a normal distribution? Extruded plastic rods are automatically cut into length 5 inches. Actual lengths are normally distributed about a mean of 5 inches and their standard deviation is 0.05 inches. (i) What proportion of rods exceed tolerance limits of 4.9 inches to 5.1 inches? (ii) Proportion of rods having tolerance rod which is greater than 6.5 inches.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Differentiate between correlation and regression analysis. Raw materials used in the production of synthetic fiber are stored in a place which has no humidity control. Measurements of the relative humidity in the storage place and the moisture content of a sample of the raw materials (both in percentage) on 10 days yielded the following results.

Humidity % (x)	Moisture content % (y)
48	53
50	44
36	31
62	16
24	8
42	25
50	11
11	13
10	12
29	16

(i) Compute correlation coefficient between humidity and moisture content and interpret the result. (ii) Find the regression equation of moisture content on humidity. (iii) Estimate the moisture content if humidity is 45%. (iv) Interpret the value of regression coefficient.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. What is sampling? Explain the main purpose of sampling. Describe briefly Stratified sampling.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What do understand by measurements of dispersion? Two batsmen A and B made the following runs in a series of cricket matches.
A: 19 25 10 30 18 28 50 33 28
B: 5 7 5 8 0 17 38 40 9 0 5 5
Who is more consistent player? Give your statistical reasoning.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define skewness and kurtosis. The first four moments about the mean are 0, 14.75, 39.75 and 152.31. Compute skewness and kurtosis and interpret the results.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What are the partition values? From the following distribution of scores 200 students of a college:
Scores: 30-40 40-50 50-60 60-70 70-80 80-90
Number of students: 14 50 60 45 20 11
Compute (i) the minimum scores obtained by top 10% students (ii) the range of middle 60% students.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define mutually exclusive events and independent events in probability. A problem of mathematics is given to three students, A, B and C whose chances of solving the problem are in ratio 2: 3: 5. Find the probability that (i) all of them solve the problem. (ii) none of them solve the problem (iii) the problem will be solved.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Define Baye's theorem. Stores A, B and C have 100, 75 and 50 employees and, respectively, 70.60 and 50 percent of these are women. Registration are equally likely among all employees regardless of sex. One employee resigns, and this is woman. What is the probability that she works in store B?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Under what conditions binomial probability distribution is used? Five unbiased coins are tossed 100 times and the following results were observed:
No. of heads: 0 1 2 3 4 5
Frequency: 5 24 35 22 10 4
Fit the Binomial distribution.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define Poisson probability distribution. Cars arrive at a petrol station at an average rate of 3 per minute. Assuming that the cars arrive at random, find the probability that (a) no cars arrive during a particular minute (b) at least one car arrive during a particular minute (c) four cars arriving per 2 minute.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Let X and Y be two continuous random variable having joint pdf $f(x,y) = c(x^2 + y^2)$, $0 \leq x \leq 1, 0 \leq y \leq 1$, otherwise 0. Determine (a) the value of c (b) $P(x < 0.5, y > 0.5)$. OR Write short notes on any two : (a) Random variable and probability distribution (b) Five number summary (c) Sampling error.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.